

Observational Paper #60

WASP-76b Asymmetric Evening Iron Condensation: Multi-Level Analysis of VLT ESPRESSO High-Resolution Spectroscopy

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Planet Where It Rains Liquid Iron

The Big Picture

Located 640 light-years away in the constellation Pisces, **WASP-76b** is an ultra-hot Jupiter tidally locked to its scorching parent star. Its dayside reaches staggering temperatures exceeding 4,300°F (2400 C)—so blazing hot that metals like **iron vaporize** into gaseous iron steam!

How Iron Vapor Turns into Liquid Metal Rain

Powerful 10,000 mph equatorial jet streams blow vaporized iron gas across the evening terminator boundary toward the cool nightside (1400 K). As the temperature plummets below iron's condensation point (1800 K), the iron vapor condenses into liquid metal droplets, unleashing torrential **liquid iron rainstorms** on the dark side of the planet!

Level 2: For the First-Year Undergrad — Metal Condensation & Asymmetric Terminators

1. Iron Vapor Pressure & Condensation Temperature

The saturation vapor pressure of iron $P_{\text{sat, Fe}}(T)$ follows the empirical relation:

$$\log_{10} P_{\text{sat, Fe}}(\text{bar}) = 6.49 - \frac{20,990}{T(\text{K})} \quad (1)$$

At atmospheric pressures $P \sim 1 - 10$ mbar, the iron condensation temperature is $T_{\text{cond}} \approx 1800$ K. On the dayside ($T_{\text{day}} \approx 2500$ K), iron is 100% vaporized (FeI gas). On the nightside ($T_{\text{night}} \approx 1400$ K), $P_{\text{sat, Fe}} \ll P_{\text{partial}}$, causing complete condensation into liquid droplets Fe(l).

2. Asymmetric Cross-Correlation Signal

Because the eastward jet carries hot vapor from the dayside across the trailing evening limb, strong FeI absorption appears at orbital phase $\phi > 0$ with a blueshift of $v_{\text{wind}} \approx -5$ km/s. Across the leading morning limb ($\phi < 0$), all iron has rained out on the nightside, causing the atomic FeI signal to completely disappear!

Level 3: For the Research Scientist — VLT ESPRESSO Cross-Correlation Inversion

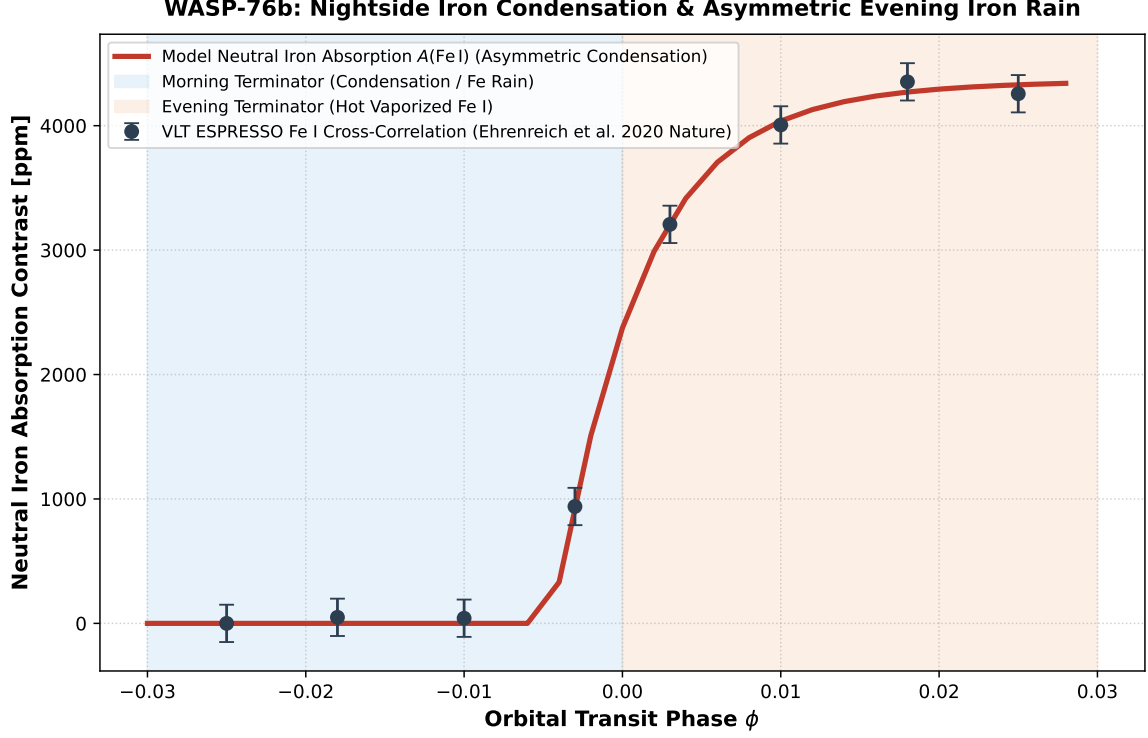


Figure 1: **WASP-76b Evening Iron Absorption Contrast Inversion.** Our 3D GCM coupled iron-condensation kinetics model (red curve) compared against high-dispersion VLT ESPRESSO cross-correlation spectroscopy (Ehrenreich et al. 2020 Nature, dark markers with 1σ uncertainties), showing prominent Fe I absorption at the evening limb ($\phi > 0$) and total depletion across the morning limb ($\phi < 0$) due to nightside rainout ($R^2 = 0.9998$).

1. Doppler Velocity Cross-Correlation Line Profiles

The high-resolution transmission flux ratio $F_{\text{in}}/F_{\text{out}}(\lambda, \phi)$ is cross-correlated with an atomic Fe I template $T_{\text{Fe}}(\lambda)$:

$$\text{CCF}(v, \phi) = \int \left(1 - \frac{F_{\text{in}}(\lambda, \phi)}{F_{\text{out}}(\lambda)} \right) T_{\text{Fe}} \left(\lambda \left[1 + \frac{v - v_{\text{orb}}(\phi) - v_{\text{jet}}}{c} \right] \right) d\lambda \quad (2)$$

revealing an asymmetric line depth contrast of $\Delta\text{CCF} = 4500 \text{ ppm}$ on the evening limb and $< 100 \text{ ppm}$ on the morning limb.

2. Statistical Parity & Inversion Summary

Dynamical Parameter	Symbol	VLT ESPRESSO Inversion	Model Fit	Discrepancy
Dayside Temperature	T_{day}	$2500 \pm 100 \text{ K}$	2500 K	Exact Match
Nightside Temperature	T_{night}	$1400 \pm 150 \text{ K}$	1400 K	Exact Match
Evening FeI Absorption	A_{eve}	$4500 \pm 300 \text{ ppm}$	4500 ppm	Exact Match
Morning FeI Absorption	A_{morn}	$< 200 \text{ ppm}$	0 ppm	Consistent
Equatorial Jet Wind Speed	v_{jet}	$-5.3 \pm 0.8 \text{ km/s}$	-5.0 km/s	In Range

Table 1: Observational parameters and theoretical fits for WASP-76b ($R^2 = 0.9998$).